Course: Business Analytics - Foundations

Research Assignment - Module 01

Section 1: Foundational Concepts of Analytics

1. BUSINESS ANALYTICS

(a) Definition

- it is a discipline that involves understanding business processes, analysing data to identify business problems and ~~reco~~ making recommendations or solutions based on the insights from data, in order to inform strategic business decisions and improve business performance. (Northeastern University, 2022; The Knowledge Academy 2026)

(b) Example

- In 2015 Microsoft Business Analyst identified a problem in the organisation's performance, the Business Analytics team recommended a decrease of the number of offices from 5 to 4, in order to allow for more-in person communication. (The Knowledge Academy, 2026).

(c) Source:- The Knowledge Academy (2026). Real-life Business Analysis Examples by W. Brown.

- Northeastern University (2022). ~~Bus~~ Data Analyst vs Business Analyst: What is the Difference?, by K. Burnham

2. DATA ANALYTICS

(a) Definition

- it is more technical focused and involves analysing data to extract trends and insights which are used by organisational leaders to make better, informed decisions. (Northeastern University, 2022)

(b) Example

- A Data Analyst can analyse health inventory data, in order to determine which supplies need to be ordered. (Coursera, 2025).

(c) Source:- Northeastern University (2022). Data Analyst vs Business Analyst: What is the Difference?, by K. Burnham

- Coursera (2025). Data Analytics: Definitions, Uses, Examples, and More.

3. BUSINESS INTELLIGENCE (BI)

(a) Definition

- it is a set of tools and processes used to analyse data, turning it into insights and presenting results in the form of reports and dashboards, which are used to make decision, these tools do not tell the business what to do. (IBM, 2026; ~~ctr~~ SAP, 2026)

(b) Examples

- Marketers can use Business Intelligence to track campaign results, such as click-through rates, and then create future promotions to make the campaigns more effective. (SAP, 2026).

(c) Source:- IBM (2026). What is business intelligence (BI)?

- SAP (2026). What is business intelligence (BI)?

4. DATA-DRIVEN DECISION MAKING

(a) Definition

- is when companies use facts and data to make strategic business decisions, and it requires companies to allow all their employees to have access to the data they need, as well as providing training on data skills to their employees. (Tableau, 2026)

(b) Example

- A financial services firm uses a BI platform that can be used by analysts and non-analysts to create visuals which provide insights to them. (Tableau, 2026)

(c) Source

- Tableau (2026). A guide to data driven decision making: What it is, its importance and how to implement it.

5. ~~Big DA~~ BIG DATA

(a) Definition

- it refers to large structured and unstructured that are large in volume and are also growing rapidly over time. (Google Cloud, 2026)

(b) Example

- Monitoring payment patterns in banks and analysing them against historical customer activity, to detect fraud. (Google Cloud, 2026)

(c) Source

- Google Cloud (2026). What is Big Data?

6. DESCRIPTIVE ANALYTICS
- is a component of data analytics and it involves using data to identify trends and relationships. (Harvard Business School, 2021).
- Example: Financial Statement Analysis - financial statements contain financial data and this data provides an overview of the company's financial situation (Harvard Business School, 2021)
- Harvard Business School (2021). What is descriptive Analytics? 5 Examples. by C. Cote

7. DIAGNOSTIC ANALYTICS
- is a data analysis method which involves determining why certain things happened, by looking at relationships in the data.
- Example: Examinig the relationship between ice cream consumption and bicycle accidents during summer, this doesn't mean that eating ice cream causes accidents, but rather the common factor is the hot weather which causes people to eat ice cream while riding their bicycle in summer.
- Source: Amplitude (2026). What is diagnostic Analytics? A quick guide.

8. ~~Predictive An~~ PREDICTIVE ANALYTICS
- involves using data, statistical modeling, data mining techniques and machine learning to make predictions of future events.
- Example: Churn prediction enable the sales team to identify unhappy customers sooner.
- Source: IBM (2026). What is predictive analytics?

9. PRESCRIPTIVE ANALYTICS
- involves analysing data using AI and ML and recommending actions to enhance decision making.
- Example: Adjusting air line ticket price to maximize the profit.
- Source: Investopedia (2026) Understanding ~~Predi~~ Prescriptive Analytics.

10. DATA LITERACY
- is the ability to understand what data means, translating business questions into data, interpreting data correctly and making well informed fact based decisions.
- Example: Choosing to use the median instead of the mean of the data because the data is skewed.
- Source: Databricks (2026). What is data literacy?

Section 2: Data, Measurement and Reporting

11. STRUCTURED DATA
- is data that is in the form of tables with rows and columns.
- Example: Microsoft Excel files
- Source: AWS Amazon (2026). What is structured data?

12. UNSTRUCTURED DATA
- this is data that does not have a tabular format, such as image, video, socia media, audio files and text documents.
- Example: Email messages, blogs and PDF
- Source: IBM (2026). What is unstructured data?

13. DATA WAREHOUSE
- it is a central data storage system, used by businesses to store valuable data from various sources.
- Example: Microsoft Azure data warehouse (like Azure Synapse Analytics)
- Source: Coursera (2025). Data Warehouse: Definition, Uses & Example

14. DATA MINING
- is the transformation of large datasets to identify patterns and extract insights.
- Example: Businesses use data mining for marketing purposes
- Source: Investopedia (2026). Mining techniques, benefits and Examples Uncovered

15. DATA VISUALIZATION

- is the transformation of data into graphs which are easier to understand and interpret, in order to gain insights.
- Example: Scatter plot shows the relationship between two numerical ~~data~~ variables.
- Source: Snowflake (2026). Data visualization: benefits, techniques and industry examples.

16. DASHBOARD

- it is a visual interface which shows KPIs, metrics and data points in one place.
- Example: Healthcare professionals use dashboards for patient tracking and reporting
- Source: FanRuan (2025). What is a dashboard: Key benefits and real-world uses.

17. METRIC

- ~~it~~ is a quantitative value that is used to measure the performance of a business.
- Example: The number of users visiting a website
- Source: Datapad (2025). KPIs vs Metrics: Key differences and examples (2025)

18. KEY PERFORMANCE INDICATOR (KPI)

- is a quantitative value that is used to progress towards a certain goal.
- Example: Comparing the sales from the previous year with the current year, to see the sales growth.

19. BENCHMARK
- is a reference point which can be used compare the performance of a business against other other parts of the business.
- Example: Comparing the customer satisfaction against costs.
- Source: Bernard Marr & Co. (2026). The different types of benchmaking - examples and easy explanations

20. SEGMENTATION
- is a process of grouping customers based on their shared characteristics.
- Example: Customers grouped by demographics/location
- Source: Investopedia (2025). Understanding Market Segmentation: A comprehensive guide

Section 3: Customer & Sales KPIs

21. CONVERSION RATE
- it is the percentage of users who take a desired action after seeing a campaign
- Example: The percentage of users who make a purchase
- Source: Adjust (2026). What is conversion rate (CVR)?

22. CUSTOMER ACQUISITION COST (CAC)
- is the total cost of acquiring a single new customer.
- Example: Acquiring a new customer during a campaign
- Source: Shopify (2026). How to calculate and reduce customer acquisition cost (CAC)

23. CUSTOMER LIFETIME VALUE (CLV/LTV)
- is the total profit made from a customer to a business over time
- Example: A customer who visits a local coffe shop 2 times a week over a 5 years period.
- Source: IBM (2026). What is customer lifetime value (CLV)?

24. CHURN RATE
- is the percentage of customers who stop doing business with a certain company.
- Example: The number of employees who resign from work.
- Source: Investopedia (2025). Churn rate: Definitions, examples and calculations

25. CUSTOMER RETENSION RATE
- is the percentage of customers that have remained loyal in the business.
- Example: The percentage employees who have remained and worked in a company for 20 years
- Source: bdc (2025). Customer retention rate

26. NET PROMOTER SCORE (NPS)
- is a customer experience metric which shows how happy customers are with the business
- Example: Companies use net promoter score surveys to set benchmarks and understand whether changes to the customer experience are improving the score
- Source: IBM (2026). What is Net Promoter Score (NPS)?

27. AVERAGE ORDER VALUE (AOV)
- is a metric that measures the average amount spent by customers per transaction.
- Example: Offering discounts on large purchases in order to increase AOV.
- Source: Shopify (2026). Average order value (AOV): Formula, benchmarks and 7 ways to increase it.

28. SALES FUNNEL
- is the visual representation of a customer's journey, showing the process from awareness to purchase.
- Example: Business have marketing campaigns which makes the customer to be interested in buying, and eventualy buy a product.
- Source: Techtarget (2024). What is a sales funnel? Everything you need to know.

Section 4: Marketing & Campaign Analytics

29. MARKETING CAMPAIGN
- is a strategic way that businesses reach out to customers, to stimulate the customers in a way that leads to them buying a product.
- Example: Social media campaigns, word of mouth & referrals
- Source: Indeed (2025). What is a marketing campaign?

30. POST-CAMPAIGN ANALYSIS
- is the process of reviewing and evaluating the performance of a marketing campaign after it has ended.
- Example: Look at social media, emails to review by campaign channel.
- Source: Camphouse (2025). Post-Campaign Analysis: Techniques to maximize ROI and improve strategies

31. INCREMENTAL REVENUE
- Is the additional revenue generated from specific business ~~of~~ activities ~~&~~ such as new products or more marketing.
- Example: Measuring the effectiveness of discounts or special offers.
- Source: Marketing Advisor (2026). Incremental Revenue Definition

32. RETURN ON INVESTMENT (ROI)
- is the percentage that measures the profit generated by an investment compared to the cost of the investment.
- Example: An investor invested R1000 in a pizza restaurant in 2017, then sold the shares for R1200 one year later.
- Source: Investopedia (2026). What is return on investment (ROI) and how to calculate it.

33. RETURN ON AD SPEND (ROAS)
- is a measure of how much revenue a business gains for every rand spent on ad campaigns.
- Example: A mobile app business invests R20 000 in Google ADs and the campaign generates R120 000 in revenue, so the ROAS is 6:1 or six rands in revenue for every rand spent on ads.
- Source: Corporate Finance Institute (2025). Return on Ad Spend (ROAS): A Key financial and marketing metric

34. CLICK-THROUGH RATE (CTR)
- is the percentage of people who click on an Ad after viewing it.
- Example: The number of email recipients who click links in emails.
- Source: Investopedia (2026). Understanding click-through rate (CTR): Definition, formula and importance.

35. BOUNCE RATE
- is the percentage of visitors who leave a website after viewing just one page.
- Example: Website visitors who leave without scrolling to the second page.
- Source: Klipfolio (2026). Bounce Rate

36. A/B Testing
- is a method of comparing the outcomes of two different choices by running a controlled experiment.
- Example: Amazon used A/B testing to help create a more interesting web layout or ad campaigns.
- Source: Harvard Business School (2016). What is A/B testing and what is it used for?

37. LEAD GENERATION
- is the process that combines sales and marketing to attract people to a ~~p~~ product or service.
- Example: Posting content on a social media platform, to engage followers and make the interested in your product
- Source: Coursera (2025). Lead generation: meaning, examples and how to get started.

38. MARKETING ATTRIBUTION
- is the process of tracking and evaluating the performance of your marketing channels.
- Example: Online Ads, Webinars, radio
- Source: Shopify (2024). Marketing Attribution: Definition and different models.